

RESULTS

Results

01 · ARIMA / SARIMA

model & forecast one series

Table 1. Model summary

	(1)
ar1	0.48 (0.11) [0.27, 0.68]
ar2	0.33 (0.09) [0.16, 0.50]
sar1	1.06 (0.12) [0.81, 1.30]
sar2	-0.53 (0.12) [-0.76, -0.30]
sma1	-0.70 (0.08) [-0.87, -0.54]
intercept	2.49 (0.12) [2.25, 2.73]
Num.Obs.	120
σ^2	0.16
Ljung-Box p	<.001
AIC	133.95
BIC	153.46
Log.Lik.	-59.98
Ljung-Box Q(5) = 23.82, p < .001 (lag 10, residual autocorrelation)	

arima()/auto.arima() return each coefficient with its SE (CI via confint()); they do not produce per-coefficient z/p — add lmtest::coeftest() if those are wanted. The Ljung-Box row tests residual autocorrelation jointly to lag = max(10, 2 × seasonal period), with df adjusted for the number of fitted ARMA terms (p + q + P + Q).

Table 2. Forecast

Period	Forecast	80% PI	95% PI
1	1.79	[1.27, 2.30]	[1.00, 2.58]
2	1.73	[1.15, 2.30]	[0.85, 2.60]
3	1.51	[0.87, 2.15]	[0.53, 2.49]
4	1.66	[0.98, 2.34]	[0.63, 2.70]
5	2.12	[1.34, 2.90]	[0.93, 3.31]
6	2.25	[1.43, 3.07]	[1.00, 3.50]
7	2.06	[1.20, 2.91]	[0.75, 3.37]
8	2.31	[1.43, 3.19]	[0.97, 3.66]

Figure. Forecast

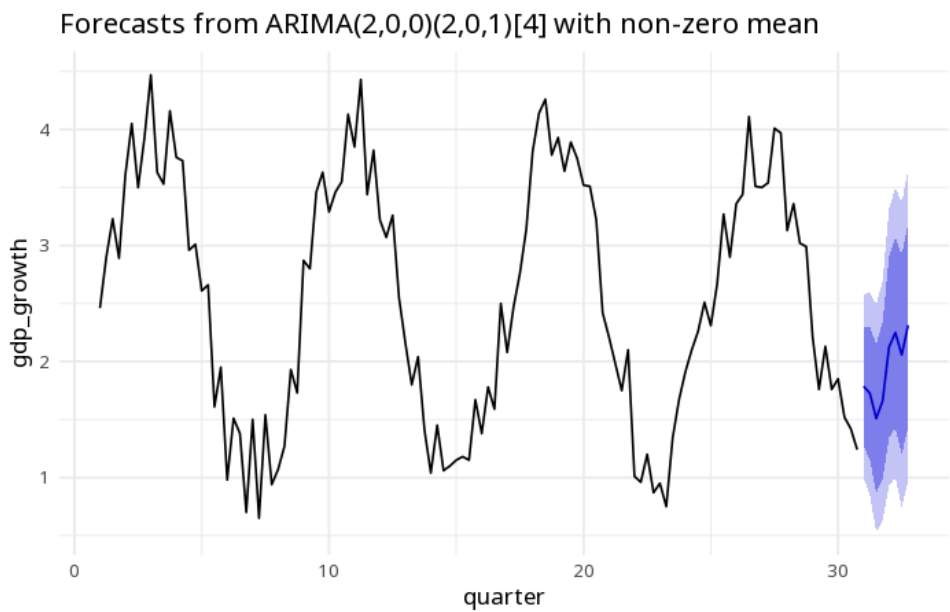
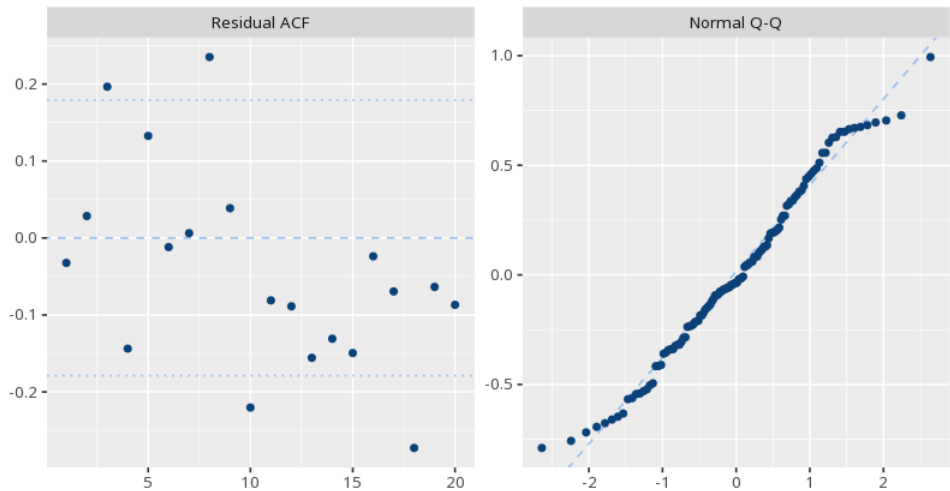


Figure. Residual diagnostics



Understanding the numbers

Estimate. The estimated value of an ARIMA/SARIMA term (an AR, MA, or seasonal coefficient), with its SE and CI. Here, the first fitted coefficient is 0.48.

σ^2 . The estimated variance of the model's residual (white-noise) innovations. Here, $\sigma^2 = 0.16$.

Ljung–Box. Tests whether the residuals still carry autocorrelation up to the reported lag - a large p means none is detected. Here, Ljung–Box $Q(5) = 23.82$, $p < .001$ (lag 10, residual autocorrelation).

AIC. Compares models fit to the same series with identical differencing and transformation - lower is better, but only relative to other such models. Here, AIC = 133.95.

BIC. Like AIC but penalizes complexity more heavily, and the same same-series/same-differencing caveat applies. Here, BIC = 153.46.

Log.Lik.. The log-likelihood: how probable the observed series is under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = -59.98 .

N. The number of observations in the series the model was fit on. Here, $N = 120$.

Period. Which future time step this forecast row refers to. Here, the first reported step is period 1.

Forecast. The model's point prediction for the series at this future period. Here, the forecast for that period is 1.79.

80% PI. An 80% prediction interval: the range expected to contain the true value 80% of the time. Here, the 80% interval is $[1.27, 2.30]$.

95% PI. A 95% prediction interval: a wider range expected to contain the true value 95% of the time. Here, the 95% interval is $[1.00, 2.58]$.

How to read this test

The chosen (p,d,q)(P,D,Q) orders describe the model. The Ljung–Box Q(df) row tests whether the residuals still carry autocorrelation up to the reported lag (a large p means none is detected); use it with AIC/BIC to judge fit. The forecast table and plot give predictions with uncertainty bands. AIC/BIC compare models only when fit to the same series with identical differencing (d, D) and transformation; lower is better, but they are relative, not absolute, measures of fit. Method: R forecast package (Hyndman & Khandakar, 2008; Hyndman & Athanasopoulos, Forecasting: Principles and Practice).

APA template: An ARIMA(2,0,0)(2,0,1[4]) model was fit (AIC=133.95); the Ljung–Box test of residual autocorrelation gave $p < .001$.

Statistical basis: ARIMA/SARIMA model selection and forecasting (Hyndman R, Athanasopoulos G, Bergmeir C, et al. (2024). forecast: Forecasting functions for time series and linear models. R package.). Full references in the exported CITATIONS.txt.

